KUET · ESE 4105 · Fuel Combustion and IC Engines

ESE 4105 Question Map

Every question from the 2020, 2021, 2022, 2023, 2024 term papers, grouped by topic and ranked by how often it comes back and how many marks it carries. Answers are in the ESE 4105 Study Pack (HTML and PDF) next to this file.

24-hour list | Pass plan | Frequency grid | Section A | Section B

24-hour priority list

You answer **3 of 4 questions in each part**, so each part has its own list. Study down both lists until the green **pass line**: about 8 h for Part A and 6 h for Part B, which leaves time for sleep and a last read of the one-shot notes. Alternate between parts (A block 1, B block 1, A block 2 …) so neither is left short. **Chance**: how often the topic came up, newer papers weigh more (2024 ×5, 2023 ×4, 2022 ×3, 2021 ×2, 2020 ×1). **Covered**: share of that part's past-paper marks you have studied once you finish the row.

Next if you have time: in Part B, the rope brake numerical (B2, 20 marks in 2024) and the Otto efficiency derivation (B7, 2024) sit below the pass line only because they appeared once, in the newest paper. Both are pure method marks and take under an hour together: they are the best extra hour. B11 (CI combustion stages) is worth reading with B13/B14, because the knock answers lean on it.

Watch the section split: this pack groups topics by subject. In 2024 cooling (A13, A14) and emissions (A16–A18) were asked in Section B, and SI combustion stages (B10) in Section A; in 2021–2023 they sat as listed here. Study by the lists, then answer whichever 3 questions in each section you know best.

Section A: Fuels, combustion, mixture preparation, cooling, lubrication, emissions

Block 1: very likely and easy · 1.7 h

Topic	Chance	Level	Covered
A1 Fuel properties: octane, cetane, API gravity, RVP, aniline point, diesel index	Very likely 100%	Easy	9%
A9 Mixture requirements: idling, cruising, full throttle, acceleration, starting	Very likely 100%	Easy	16%
A13 Need for cooling and lubrication; effects of overcooling	Very likely 93%	Easy	21%
A14 Air vs liquid cooling; water cooling system, thermostat, thermosyphon	Very likely 80%	Easy	31%

Block 2: very likely, medium effort · 3.0 h

Topic	Chance	Level	Covered
A18 Catalytic converter: construction, reactions, temperature and equivalence ratio effects	Very likely 100%	Medium	39%
A2 Proximate vs ultimate analysis; HCV vs LCV	Very likely 93%	Medium	44%
A16 Pollutant formation in SI and CI engines; NOx mechanism; smog	Very likely 87%	Medium	52%
A11 Fuel injection system: purpose, components, types	Very likely 80%	Medium	58%

Block 4: likely · 2.6 h

Topic	Chance	Level	Covered
A4 Classification of fuels; storage and handling of solid, liquid and gaseous fuels	Likely 60%	Easy	62%
A8 Carburetion and the simple carburetor; compensating jet	Likely 60%	Easy	65%
A12 Injector nozzles: types, spray pattern, atomization, sac volume	Likely 67%	Medium	69%
A10 Venturi throat diameter numerical (compressible, isentropic)	Likely 60%	Hard	72%

Only if time is left · 3.2 h

Topic	Chance	Level	Covered
A17 Particulate emission, its measurement, smoke colours	Possible 53%	Easy	76%
✔ Pass line: 76% of Section A marks covered, enough to choose 3 of the 4 questions (about 7.7 h)			
A3 Alternative fuels: ethanol, biodiesel, methanol, hydrogen	Possible 40%	Easy	83%
A15 Lubrication: parts lubricated, oil selection, classification, mist lubrication	Possible 33%	Easy	86%
A7 Producer gas air requirement numerical; fluidized bed combustion	Possible 33%	Easy	89%

A6 Combustion: definition, types, air-fuel ratio and its effect	Possible 33%	Easy	91%
A5 Petroleum refining: fractional distillation, cracking, reforming	Possible 13%	Easy	93%
A19 Emission control techniques; NOx reduction; SI vs CI emissions; blow-by	Possible 20%	Medium	100%

Section B: Performance, cycles, valve timing, SI/CI combustion, knock, supercharging

Block 2: very likely, medium effort · 1.5 h

Topic	Chance	Level	Covered
B15 Supercharging and turbocharging: effects, SI vs CI suitability, limits, modifications	Very likely 100%	Medium	12%
B1 Performance parameters and performance curves (BP, torque, BSFC, ηv vs speed)	Very likely 87%	Medium	20%

Block 4: likely · 4.7 h

Topic	Chance	Level	Covered
B6 Valve timing diagram, valve overlap, why inlet valves are larger	Likely 73%	Easy	26%
B10 Stages of combustion in SI engine	Likely 73%	Easy	31%
B5 Stoichiometry, equivalence ratio, rich vs lean; why CI runs lean	Likely 67%	Easy	37%
B12 Surface ignition, pre- vs post-ignition, end gas, flame front	Likely 60%	Easy	41%
B13 Knock in SI vs CI; ignition delay SI vs CI; SI knock limited	Likely 67%	Medium	53%
B4 Indicator diagram: indicated and pumping work; pumping loss, WOT, turbocharged engines	Likely 67%	Medium	61%
B14 CI knock factors and ignition delay; good SI fuel is bad CI fuel; poor fuel in large slow CI; smoke limited	Likely 60%	Medium	72%
B9 SI mixture formation strategies; HCCI and DISI	Likely 60%	Medium	75%

✔ Pass line: 75% of Section B marks covered, enough to choose 3 of the 4 questions (about 6.2 h)

Only if time is left · 3.7 h

Topic	Chance	Level	Covered
B8 Scavenging; two-stroke vs four-stroke; why 2-stroke CI is common but 2-stroke SI is not	Possible 40%	Easy	83%
B11 Stages of combustion in CI engine (ROHR vs crank angle); flames; SIT	Possible 53%	Medium	90%

B3 Volumetric efficiency: significance, mean piston speed, valve timing and speed effects, quasi-static effect	Possible 33%	Medium	94%
B2 Rope brake engine test numerical (BP, IP, ηm, ηbth, BSFC)	Possible 33%	Medium	97%
B7 Air-standard cycles: Otto efficiency derivation; Diesel cut-off ratio and γ	Possible 33%	Hard	100%

Pass plan

2021-2024 papers: 210 marks, 4 questions of 35 per section, answer 3 per section. 2020 was a short 120-mark paper (3 questions of 30 per section, answer 2). All materials are Mahmudul Hasan Sir's slides. The split between sections moves a little: in 2024 Section A had fuels, combustion, carburetion and injection and Section B had performance, cycles, cooling, emissions and supercharging; in 2021-2023 cooling and emissions sat in Section A and SI/CI combustion and knock in Section B. Topics here are grouped by subject: A = fuels, combustion, mixture preparation, cooling/lubrication, emissions; B = engine performance, cycles, valve timing, combustion and knock, supercharging. "Must study" topics came up in 3 or more of the 5 papers.

SECTION A: FUELS, COMBUSTION, MIXTURE PREPARATION, COOLING, LUBRICATION, EMISSIONS

13 topics

cover 84% of all Section A marks asked

- Fuel properties, proximate/ultimate, alternative fuels
- Carburetor, mixture requirements, venturi numerical
- Fuel injection system and nozzles; cooling and lubrication
- Pollutants, particulates, catalytic converter, emission control

SECTION B: PERFORMANCE, CYCLES, VALVE TIMING, SI/CI COMBUSTION, KNOCK, SUPERCHARGING

12 topics

cover 91% of all Section B marks asked

- Performance curves, ηv, pumping loss, equivalence ratio + rope brake numerical
- Valve timing, Otto/Diesel cycle, scavenging and 2-stroke
- SI and CI combustion stages, surface ignition
- Knock SI vs CI and the justify-statements; supercharging

- **Learn the two numericals first.** The venturi throat diameter (2024 3d, 2023 3d) and the rope brake test (2024 6b, 20 marks) are pure formula marks.
- **Section B is the same ten statements every year:** CI runs lean, pumping loss SI vs CI, WOT, pre-ignition worse than post-ignition, SI knock limited, CI smoke limited, good SI fuel is bad CI fuel, low-quality fuel in big slow CI, 2-stroke CI vs 2-stroke SI, which engine suits supercharging. Memorise one paragraph each.
- **Always draw:** SI p-θ (three stages), CI ROHR-θ (four stages), valve timing circle, catalytic converter efficiency vs temperature and vs ϕ, water cooling circuit, A/F vs throttle opening.
- **Section A emissions block** (pollutant formation, PM, catalytic converter, NOx control) appears every year; it gives you a safe full question.
- **"Good to have"** topics are the backup for your third question in a section.

Frequency grid

Marks asked from each topic in each paper. Darker means more marks. Click a topic to jump to its questions.

Marks per paper: 1–8 9–16 17–24 25+

Sort Section A: Best order (likely + easy) | Most likely first | Easiest first | Paper order

SECTION A: FUELS, COMBUSTION, MIXTURE PREPARATION, COOLING, LUBRICATION, EM
PER PAPER

Topic	2024	2023	2022	2021	2020	Years	Marks	
A1 Fuel properties: octane, cetane, API gravity, RVP, aniline point, diesel index	8	18	15	10	11	5/5	62	Must st
A9 Mixture requirements: idling, cruising, full throttle, acceleration, starting	9	10	10	8	14	5/5	51	Must st
A13 Need for cooling and lubrication; effects of overcooling	12	5	14	6		4/5	37	Must s
A14 Air vs liquid cooling; water cooling system, thermostat, thermosyphon	23	30		15	8	4/5	76	Must s

A18 Catalytic converter: construction, reactions, temperature and equivalence ratio effects	12	12	11	10	12	5/5	57	Must stu
A2 Proximate vs ultimate analysis; HCV vs LCV	10	6	8	12		4/5	36	Must stu
A16 Pollutant formation in SI and CI engines; NOx mechanism; smog	12	11	21		10	4/5	54	Must stu
A11 Fuel injection system: purpose, components, types	15	10	16			3/5	41	Must stu
A4 Classification of fuels; storage and handling of solid, liquid and gaseous fuels	17	11				2/5	28	Good
A8 Carburetion and the simple carburetor; compensating jet	18	8				2/5	26	Mus
A12 Injector nozzles: types, spray pattern, atomization, sac volume	10		9	12		3/5	31	Must s
A10 Venturi throat diameter numerical (compressible, isentropic)	8	7				2/5	15	Good
A17 Particulate emission, its measurement, smoke colours		12	12		8	3/5	32	Must s
A3 Alternative fuels: ethanol, biodiesel, methanol, hydrogen			12	13	25	3/5	50	Must s
A15 Lubrication: parts lubricated, oil selection, classification, mist lubrication			11	14		2/5	25	Good t
A7 Producer gas air requirement numerical; fluidized bed combustion	19					1/5	19	Skip if
A6 Combustion: definition, types, air-fuel ratio and its effect	16					1/5	16	Good t
A5 Petroleum refining: fractional distillation, cracking, reforming				15		1/5	15	Skip if

A19 Emission control techniques; NOx reduction; SI vs CI emissions; blow-by				25	22	2/5	47	Must st

Sort Section B: Best order (likely + easy) | Most likely first | Easiest first | Paper order

SECTION B: PERFORMANCE, CYCLES, VALVE TIMING, SI/CI COMBUSTION, KNOCK, SUPE

ASKED PER PAPER

Topic	2024	2023	2022	2021	2020	Years	Marks	
B15 Supercharging and turbocharging: effects, SI vs CI suitability, limits, modifications	11	19	14	7	30	5/5	81	Must stu
B1 Performance parameters and performance curves (BP, torque, BSFC, ηv vs speed)	12	14	12		9	4/5	47	Must stu
B6 Valve timing diagram, valve overlap, why inlet valves are larger	10		14	12	5	4/5	41	Mus
B10 Stages of combustion in SI engine	10	10		12		3/5	32	Mus
B5 Stoichiometry, equivalence ratio, rich vs lean; why CI runs lean		9	11	9	9	4/5	38	Mus
B12 Surface ignition, pre- vs post-ignition, end gas, flame front		7	11	11		3/5	29	Mus
B13 Knock in SI vs CI; ignition delay SI vs CI; SI knock limited		12	23	22	20	4/5	77	Must s
B4 Indicator diagram: indicated and pumping work; pumping loss, WOT, turbocharged engines		12	13	16	12	4/5	53	Must s
B14 CI knock factors and ignition delay; good SI fuel is bad CI fuel; poor fuel in large slow CI; smoke limited		34	10	25		3/5	69	Must s
B9 SI mixture formation strategies; HCCI and DISI	9	12				2/5	21	Good t
B8 Scavenging; two-stroke vs four-stroke; why 2-stroke CI is			18	14	19	3/5	51	Must s

common but 2-stroke SI is not						
B11 Stages of combustion in CI engine (ROHR vs crank angle); flames; SIT	11	24		10	3/5	45	Must st
B3 Volumetric efficiency: significance, mean piston speed, valve timing and speed effects, quasi-static effect		10	19		2/5	29	Must st
B2 Rope brake engine test numerical (BP, IP, ηm, ηbth, BSFC)	20				1/5	20	Skip if s
B7 Air-standard cycles: Otto efficiency derivation; Diesel cut-off ratio and γ	19				1/5	19	Good t

Section A: Fuels, combustion, mixture preparation, cooling, lubrication, emissions

questions 1-4 (1-3 in 2020)

Sort Section A: Best order (likely + easy) | Most likely first | Easiest first | Paper order

A1 Fuel properties: octane, cetane, API gravity, RVP, aniline point, diesel index

Must study · Very likely · 100% · Easy

Asked in 5 of 5 papers · 62 marks in total

STUDY THESE

- Octane number = % iso-octane in iso-octane + n-heptane blend that knocks like the fuel; cetane number = % n-hexadecane in hexadecane + 1-methylnaphthalene blend with the same ignition delay
- Higher octane lets an SI engine run higher compression ratio / spark advance without knock, so more power and efficiency
- API = 141.5/SG − 131.5 (hydrometer); RVP at 38 °C in a 4:1 air chamber (volatility); aniline point (aromatics); Diesel index = aniline point (°F) × API / 100

> 2024 Q1(c) 8 marks
>
> Explain octane and cetane number. How does the octane number correlate with IC engine performance?

> 2023 Q1(a) 12 marks
>
> Briefly explain the measurement methods and significance of: (i) API gravity, (ii) Reid vapor pressure, (iii) Cetane number, (iv) Octane number.

> 2023 Q1(c) 6 marks
>
> How does the octane number correlate with the performance of IC engines?

> 2022 Q1(a) 15 marks
>
> Briefly explain the measurement methods and significance of: (i) API Gravity, (ii) Reid Vapor Pressure, (iii) Aniline Point and Diesel Index.

> 2021 Q1(b) 10 marks
>
> What is octane and cetane number? Explain what it means by a cetane rating of 60 and an octane rating of 85?

> 2020 Q1(a) 11 marks
>
> How can you use hydrogen as a fuel in CI engine? Explain the physical significances of octane number and cetane number with example.

Full answer: ESE 4105 Study Pack, topic A1.

A9 Mixture requirements: idling, cruising, full throttle, acceleration, starting

Must study · Very likely · 100% · Easy

Asked in 5 of 5 papers · 51 marks in total

STUDY THESE

- Three ranges (A/F vs throttle curve): idling rich (~12), cruising lean best economy (~16-17), full power rich best power (~12.5-13)
- Transient: cold start very rich (choke), acceleration extra squirt (accelerator pump)
- Why dilute the mixture (excess air/EGR): more expansion work, less pumping work, less heat loss, lower NOx

> 2024 Q3(b) 9 marks
>
> How does the air-fuel mixture change with varying engine speeds from idling to high-speed operation?

> 2023 Q3(b) 10 marks

> How does the air-fuel mixture requirement change between idling, cruising, and full throttle conditions in IC engine?

> 2022 Q2(c) 10 marks
> Why it is advantageous to dilute fuel-air mixture? How quasi-static effect influences the volumetric efficiency of an internal combustion engine.

> 2021 Q3(c) 8 marks
> Explain the variation in mixture quality required by a SI engine running with different operating conditions.

> 2020 Q1(b) 14 marks
> Why do you need extra fuel when the throttle is opened? (second part)

Full answer: ESE 4105 Study Pack, topic A9.

A13 Need for cooling and lubrication; effects of overcooling

Must study · Very likely · 93% · Easy

Asked in 4 of 5 papers · 37 marks in total

STUDY THESE

- Gas temperature ~2500 °C; ~30 % of fuel energy goes to coolant; keep wall ~180-200 °C
- Without cooling: oil film breaks, piston seizure, thermal stress, knock/pre-ignition, low volumetric efficiency
- Overcooling: poor vaporization, more heat loss, lower thermal efficiency, oil dilution and sludge, corrosion, more friction

> 2024 Q7(a) 12 marks
> Why cooling and lubrication systems are necessary for IC engines?

> 2023 Q2(a) 5 marks
> Why cooling and lubricating systems are necessary for IC engines?

> 2022 Q2(a) 14 marks
> What are the effects of overcooling the IC engines? What are the advantages and disadvantages of different cooling systems?

> 2021 Q2(a) 6 marks
> Why cooling and lubrication systems are necessary for IC engines?

Full answer: ESE 4105 Study Pack, topic A13.

A14 Air vs liquid cooling; water cooling system, thermostat, thermosyphon

Must study · Very likely · 80% · Easy

Asked in 4 of 5 papers · 76 marks in total

STUDY THESE

- Air cooling: fins, small engines, aircraft, simple, light; liquid cooling: jackets + radiator + pump + thermostat, cars/large engines, uniform
- Water cooling circuit sketch: jacket → thermostat → top tank → radiator core (fan) → bottom tank → pump → jacket
- Thermostat (wax/bellows) closed below ~70-80 °C; stuck closed → overheating; stuck open → overcooling. Thermosyphon: density difference, slow, stops if level falls

> 2024 Q7(b) 11 marks
>
> How does a liquid cooling system differ from an air cooling system in terms of applications in IC engines?

> 2024 Q7(c) 12 marks
>
> Describe the water cooling system of IC engine with a schematic diagram.

> 2023 Q2(b) 10 marks
>
> How does a liquid cooling system differ from an air cooling system in terms of applications in IC engines?

> 2023 Q2(d) 10 marks
>
> Describe the water cooling system of IC engine with a schematic diagram.

> 2023 Q2(c) 10 marks
>
> How does the use of a thermostat in a water cooling system regulate engine temperature and what happens if the thermostat fails to function properly?

> 2021 Q3(a) 15 marks
>
> Describe the water cooling system of IC engine with a schematic diagram.

> 2020 Q2(c) 8 marks
>
> How is the circulation achieved in thermo-syphon system? What are the limitations of this system?

Full answer: ESE 4105 Study Pack, topic A14.

A18 Catalytic converter: construction, reactions, temperature and equivalence ratio effects

Must study · Very likely · 100% · Medium

Asked in 5 of 5 papers · 57 marks in total

STUDY THESE

- Stainless shell, ceramic (cordierite) honeycomb, alumina wash-coat, Pt/Pd (oxidation) + Rh (NO reduction), CeO_2 oxygen storage
- $2CO + O_2 \rightarrow 2CO_2$; $HC + O_2 \rightarrow CO_2 + H_2O$; $2NO + 2CO \rightarrow N_2 + 2CO_2$
- Light-off 250-300 °C (50 % efficiency); 3-way works only in a narrow window at $\phi \approx 1$ (lean kills NOx conversion, rich kills CO/HC)

> 2024 Q8(b) 12 marks
>
> What is a catalytic converter? Explain how the effect of temperature and equivalence ratio on the conversion efficiency of catalytic converter.

> 2023 Q4(c) 12 marks
>
> How do catalytic converter in IC engines reduce the emissions of HC, CO and NOx, and what are the chemical reactions involved?

> 2022 Q4(b) 11 marks
>
> With proper sketch, discuss the construction and operation of a catalytic converter.

> 2021 Q4(a) 10 marks
>
> How temperature and fuel equivalence ratio affect the conversion efficiency of a catalytic converter? Explain with necessary graphs.

> 2020 Q2(a) 12 marks
>
> What is catalytic converter? With a schematic diagram explain how a 3-way catalytic converter does treat exhaust from engine?

Full answer: ESE 4105 Study Pack, topic A18.

A2 Proximate vs ultimate analysis; HCV vs LCV

Must study · Very likely · 93% · Medium

Asked in 4 of 5 papers · 36 marks in total

STUDY THESE

- Proximate: moisture (108-110 °C), volatile matter (950 °C, 7 min, covered), ash (725 °C, open), fixed carbon by difference
- Ultimate: C, H, N, S, O and ash by weight; gives theoretical air, CV (Dulong) and flue gas composition
- HCV counts latent heat of the water formed; LCV = HCV − 9H × 2442 kJ/kg (water leaves as vapour)

> 2024 Q1(d) 10 marks
>
> Explain how ultimate analysis contributes to understanding the combustion behavior and energy potential of a fuel.

> 2023 Q1(b) 6 marks
>
> What are the key differences between proximate analysis and ultimate analysis of fuel?

> 2022 Q1(b) 8 marks
>
> Compare and contrast ultimate analysis and proximate analysis.

> 2021 Q1(c) 12 marks
>
> Explain the differences between: (i) Ultimate analysis and proximate analysis, (ii) Higher calorific value and lower calorific value.

Full answer: ESE 4105 Study Pack, topic A2.

A16 Pollutant formation in SI and CI engines; NOx mechanism; smog

Must study · Very likely · 87% · Medium

Asked in 4 of 5 papers · 54 marks in total

STUDY THESE

- SI: HC from crevices, quench, oil layer absorption/desorption, deposits, incomplete combustion; CO from rich mixture; NOx thermal
- CI: lean zone HC, sac volume HC, soot in rich spray core, NOx in near-stoichiometric diffusion flame
- Zeldovich: $O + N_2 \rightleftharpoons NO + N$, $N + O_2 \rightleftharpoons NO + O$, $N + OH \rightleftharpoons NO + H$; strongly temperature and residence-time dependent (above 1800 K)

> 2024 Q8(a) 12 marks

Explain the pollutant formation mechanism in CI Engine.

2023 Q4(a) 11 marks

Describe the pollutants formation mechanisms in SI engine with appropriate diagrams.

2022 Q4(a) 12 marks

What is smog? Discuss the mechanisms of NOx formation? How do reduce this emission?

2022 Q3(c) 9 marks

What is the relationship between NOx formation and combustion time? (second part)

2020 Q3(a) 10 marks

What is meant by total emission control packages? How NOx and PM are generated in CI engines?

Full answer: ESE 4105 Study Pack, topic A16.

A11 Fuel injection system: purpose, components, types

Must study · Very likely · 80% · Medium

Asked in 3 of 5 papers · 41 marks in total

STUDY THESE

- Purpose: meter, time, inject at the correct rate, atomize and distribute fuel equally; cut off sharply (no dribble)
- Components: tank, feed pump, filters, high-pressure (injection) pump, governor, HP lines, injector/nozzle
- Types: air injection (old) vs solid (airless) injection: individual pump, unit pump/injector, distributor, common rail; SI: throttle-body, port (MPFI), GDI

2024 Q4(a) 7 marks

What is the purpose of a fuel injection system in an IC engine?

2024 Q4(b) 8 marks

Describe the main components of a fuel injection system and their functions.

2023 Q3(c) 10 marks

Describe the different types of fuel injection systems and injector nozzles.

2022 Q3(a) 16 marks

> Using proper sketch, discuss various fuel injection systems and injector nozzles. Compare their relative merits and demerits.

Full answer: ESE 4105 Study Pack, topic A11.

A4 Classification of fuels; storage and handling of solid, liquid and gaseous fuels

Good to have · Likely · 60% · Easy

Asked in 2 of 5 papers · 28 marks in total

STUDY THESE

- By origin: primary (natural: wood, coal, crude, natural gas) vs secondary (derived: coke, petrol, LPG, producer gas); by state: solid / liquid / gas
- Solid: spontaneous combustion, dust explosion, weathering → covered, ventilated, compacted heaps, temperature checks
- Liquid: flash point, vapour, leaks → tanks with vents, bunds, earthing; Gas: high pressure / cryogenic cylinders, leak detection, ventilation, flammability limits

> 2024 Q1(a) 8 marks
>
> What are the primary classifications of fuels, and how are they differentiated based on their origin?

> 2024 Q1(b) 9 marks
>
> What factors influence the safe storage and handling of solid, liquid, and gaseous fuels?

> 2023 Q1(d) 11 marks
>
> How do the storage and handling requirement of solid, liquid and gaseous fuels differ and what safety considerations are associated with each?

Full answer: ESE 4105 Study Pack, topic A4.

A8 Carburetion and the simple carburetor; compensating jet

Must study · Likely · 60% · Easy

Asked in 2 of 5 papers · 26 marks in total

STUDY THESE

- Carburetion = forming a combustible air-fuel mixture outside the cylinder (metering, atomizing, vaporizing, distributing)

- Venturi principle: velocity ↑, pressure ↓ at throat; float chamber at atmospheric pressure pushes fuel out of the jet
- Simple carburetor richens as air flow rises; compensating jet / air bleed / emulsion tube keep A/F constant

> 2024 Q3(a) 8 marks
>
> Briefly explain the working principle of a carburetor in IC engines with neat sketches.

> 2024 Q4(d) 10 marks
>
> Explain the function of the compensating jet in a carburetor and how it helps maintain proper combustion.

> 2023 Q3(a) 8 marks
>
> What is carburation in an IC engine? Describe the working principle of a carburator in IC engine.

Full answer: ESE 4105 Study Pack, topic A8.

A12 Injector nozzles: types, spray pattern, atomization, sac volume

Must study · Likely · 67% · Medium

Asked in 3 of 5 papers · 31 marks in total

STUDY THESE

- Single-hole (narrow 15° spray, high pressure, dribbles), multi-hole (4-18 holes of 0.15-0.35 mm, open chambers), pintle (hollow cone 0-60°, swirl/pre-chamber), pintaux (auxiliary hole, cold start)
- Atomization by high ΔP across small orifice; penetration vs cone angle vs droplet size set by hole size, L/d, pressure
- Sac volume: small space below needle seat; fuel left there evaporates late → HC emission; VCO nozzles remove it

> 2024 Q4(c) 10 marks
>
> Explain how fuel injection nozzles control the spray pattern and atomization of fuel for efficient combustion.

> 2022 Q3(c) 9 marks
>
> What is sac volume? What is the relationship between NOx formation and combustion time?

> 2021 Q3(b) 12 marks

> Describe the different types of injection nozzles with their relative merits and demerits.

Full answer: ESE 4105 Study Pack, topic A12.

A10 Venturi throat diameter numerical (compressible, isentropic)

Good to have · Likely · 60% · Hard

Asked in 2 of 5 papers · 15 marks in total

STUDY THESE

- $C_2 = C_v \sqrt{(2 c_p T_1 [1 - (p_2/p_1)^{((\gamma-1)/\gamma)}])}$ → solve p_2
- $\rho_1 = p_1/RT_1$, $\rho_2 = \rho_1 (p_2/p_1)^{(1/\gamma)}$
- $A_2 = \dot{m} / (\rho_2 C_2)$, $d = \sqrt{(4A_2/\pi)}$; answers ≈ 30-35 mm

> 2024 Q3(d) 8 marks
> An engine requires 6 kg of air per minute from its carburetor. The air is at a pressure of 1.02 bar and a temperature of 25 °C. Determine the throat diameter of the venturi when the air flow velocity at the throat is 100 m/s with a velocity coefficient of 0.75. Assume isentropic flow and consider air as compressible. R = 287 J/kg·K.

> 2023 Q3(d) 7 marks
> An engine requires 5 kg of air per minute from its carburator. The air is at a pressure of 1.05 bar and a temperature of 25 °C. Determine the throat diameter of the venturi when the air-flow velocity at the throat is 1.02 m/s [sic] with a velocity coefficient of 0.8. Assume isentropic flow and consider air as compressible.

Full answer: ESE 4105 Study Pack, topic A10.

A17 Particulate emission, its measurement, smoke colours

Must study · Possible · 53% · Easy

Asked in 3 of 5 papers · 32 marks in total

STUDY THESE

- PM = everything collected on a filter from diluted exhaust at ≤ 325 K: soot (carbon) + soluble organic fraction
- Measurement: dilution tunnel + filter weighing, smoke meters (Bosch filter paper, Hartridge opacity), particle counters

- CI ≫ SI because heterogeneous combustion has rich spray zones; smoke colour: black = soot, blue = lube oil, white = unburnt fuel / water

> 2023 Q4(b) 12 marks
>
> What is particulate emission? How can particulate emission be measured? Why does particulate emission from CI engines differ significantly than that of SI engines?

> 2022 Q4(c) 12 marks
>
> What is particulate emission? How particulate emission can be measured? Why does particulate emission from CI engines differ significantly than that of SI engines?

> 2020 Q3(c) 8 marks
>
> What is smoke? "Smoke colors are indicative of dominant source of particulate" – Explain.

Full answer: ESE 4105 Study Pack, topic A17.

A3 Alternative fuels: ethanol, biodiesel, methanol, hydrogen

Must study · Possible · 40% · Easy

Asked in 3 of 5 papers · 50 marks in total

STUDY THESE

- Ethanol: high octane (~108), E10/E85, lower heating value, renewable; Biodiesel (FAME) by transesterification, higher cetane, lower CO/HC/soot, slightly more NOx
- Methanol problems: half the heating value, toxic, corrosive, cold-start difficulty, high latent heat, formaldehyde emission → extra fuel / enrichment needed
- Hydrogen in CI: very high self-ignition temperature (~585 °C) so it cannot self-ignite; use dual-fuel (diesel pilot), glow plug or spark assist

> 2022 Q1(c) 12 marks
>
> Describe the possible pathways of using alcohol based fuels and hydrogen fuel for IC engines.

> 2021 Q1(a) 13 marks
>
> What is meant by "Alternative fuels"? Discuss the prospects of ethanol and biodiesel as a fuel for IC engines.

> 2020 Q1(a) 11 marks

> How can you use hydrogen as a fuel in CI Engine? (first part)

> 2020 Q1(b) 14 marks
> What are the problems of methanol as alternative fuel? Why do you need extra fuel when the throttle is opened?

Full answer: ESE 4105 Study Pack, topic A3.

A15 Lubrication: parts lubricated, oil selection, classification, mist lubrication

Good to have · Possible · 33% · Easy

Asked in 2 of 5 papers · 25 marks in total

STUDY THESE

- Parts: cylinder walls and piston/rings, piston pin, main and big-end bearings, camshaft bearings, valve gear, timing gears, pump, fan
- SAE grades: single (5W-20W winter at −18 °C; 20-50 at 99 °C) and multigrade 20W-50; select by viscosity index, pour point, flash point
- Mist (petroil) lubrication in 2-stroke: 2-3 % oil in petrol (50:1), no separate system, but smoke and deposits

> 2022 Q2(b) 11 marks
> What are the various components to be lubricated in an engine? Write the selection criteria of lubricating oils in various weather conditions.

> 2021 Q2(b) 14 marks
> How lubricating oils are classified? Discuss the mist lubrication system with diagram.

Full answer: ESE 4105 Study Pack, topic A15.

A7 Producer gas air requirement numerical; fluidized bed combustion

Skip if short · Possible · 33% · Easy

Asked in 1 of 5 papers · 19 marks in total

STUDY THESE

- Per m^3 of gas: $O_2 = 0.5\,H_2 + 0.5\,CO + 2\,CH_4 - O_2$ in gas; air = O_2 / 0.21

- Bubbling FBC: sand/limestone bed fluidised by air, 750-950 °C, biomass devolatilises then char burns
- Heat transfer: bed to immersed tubes by particle convection + gas convection + radiation (very high h)

> 2024 Q2(d) 10 marks
>
> A producer gas used as a fuel has the volumetric composition H_2 25 %, CO 15 %, CH_4 5 %, CO_2 15 % and N_2 40 %. Find the volume of air required for complete combustion of 1 m^3 of the gas. Air contains 21 % by volume of oxygen.

> 2024 Q2(c) 9 marks
>
> Explain the combustion mechanism and heat transfer process involved in bubbling fluidized bed combustion of solid biomass.

Full answer: ESE 4105 Study Pack, topic A7.

A6 Combustion: definition, types, air-fuel ratio and its effect

Good to have · Possible · 33% · Easy

Asked in 1 of 5 papers · 16 marks in total

STUDY THESE

- Combustion = rapid exothermic oxidation of fuel; needs fuel + oxygen + ignition temperature (fire triangle)
- Types: spontaneous, rapid, complete, incomplete, explosion; premixed vs diffusion
- A/F ratio (mass air / mass fuel): stoichiometric ~14.7 for petrol; lean = efficient but slow; rich = power but CO, HC, soot

> 2024 Q2(a) 7 marks
>
> What is combustion? Briefly explain the different types of combustion.

> 2024 Q2(b) 9 marks
>
> Explain the concept of air-fuel ratio and describe how it affects the efficiency and completeness of combustion.

Full answer: ESE 4105 Study Pack, topic A6.

A5 Petroleum refining: fractional distillation, cracking, reforming

Skip if short · Possible · 13% · Easy

Asked in 1 of 5 papers · 15 marks in total

STUDY THESE

- Crude is a mixture of hydrocarbons with impurities, so it must be separated and upgraded
- Fractional distillation: heat to 300-350 °C, column 20-30 m with bubble caps, fractions every ~50 °C
- Cracking breaks heavy molecules; reforming (thermal 500-600 °C or catalytic Pt 480-540 °C) raises gasoline octane

> 2021 Q2(c) 15 marks
>
> Why petroleum refining is necessary? Explain fractional distillation and reforming process.

Full answer: ESE 4105 Study Pack, topic A5.

A19 Emission control techniques; NOx reduction; SI vs CI emissions; blow-by

Must study · Possible · 20% · Medium

Asked in 2 of 5 papers · 47 marks in total

STUDY THESE

- Three routes: engine modification (design + operating), exhaust after-treatment (thermal reactor, catalytic converter, EGR, DPF, SCR), fuel modification
- NOx down by lowering peak temperature and O_2: EGR, retarded timing, water injection, lean/rich, SCR with urea
- Blow-by = gas leaking past rings into crankcase; PCV valve routes it back to intake. SI emits more CO and HC; CI more NOx-PM

> 2021 Q4(c) 13 marks
>
> What are the common emission control techniques? Discuss the strategies to reduce NOx emission from IC engines?

> 2021 Q4(b) 12 marks
>
> Compare the emissions from SI engine with that from CI engine.

> 2020 Q3(b) 12 marks
>
> What is blow-by loss? How can you achieve effective removal of blow by? Explain with schematic diagram.

2020 Q3(a) 10 marks

What is meant by total emission control packages? (first part)

Full answer: ESE 4105 Study Pack, topic A19.

Section B: Performance, cycles, valve timing, SI/CI combustion, knock, supercharging

questions 5-8 (4-6 in 2020)

Sort Section B: Best order (likely + easy) | Most likely first | Easiest first | Paper order

B15 Supercharging and turbocharging: effects, SI vs CI suitability, limits, modifications

Must study · Very likely · 100% · Medium

Asked in 5 of 5 papers · 81 marks in total

STUDY THESE

- Supercharging = supplying air above atmospheric density (mechanically driven); turbocharging uses an exhaust turbine
- Effects: power ↑ 30-50 %, better mechanical efficiency, better combustion; but higher p and T, knock (SI), thermal load
- CI is more suitable (boost reduces delay and knock); SI limited by knock. Modifications: lower r, stronger parts, more cooling, bigger valve overlap, larger pump/nozzles, intercooler

2024 Q8(c) 11 marks

Explain the concept of supercharging and how it affects engine power and efficiency.

2023 Q8(a) 11 marks

Describe the effect of supercharging. Which engine is more suitable for supercharging among SI engine or CI engine? Explain why?

2023 Q8(b) 8 marks

What types of engine modifications are required if supercharging is to be employed? Explain.

> 2022 Q8(a) 14 marks
>
> Discuss the limits of supercharging in both SI engine and CI engine.

> 2021 Q6(d) 7 marks
>
> What are the benefits of supercharging? (first part)

> 2020 Q6(b) 10 marks
>
> Describe the effects of supercharging. Which engine is more suitable for supercharging – SI engine or CI engine? Why?

> 2020 Q6(a) 10 marks
>
> Discuss the supercharging limits for both SI engine and CI engine.

> 2020 Q6(c) 10 marks
>
> How can you modify an engine for supercharging? What are the limitations of turbocharging?

Full answer: ESE 4105 Study Pack, topic B15.

B1 Performance parameters and performance curves (BP, torque, BSFC, ηv vs speed)

Must study · Very likely · 87% · Medium

Asked in 4 of 5 papers · 47 marks in total

STUDY THESE

- Definitions: ηv = actual air mass / (ρ_a V_s); BSFC = ṁ_f / BP; IMEP = indicated work / V_s; BP = 2πNT/60
- Torque and ηv peak at mid speed; BP rises to a maximum at high speed then falls (friction ↑, ηv ↓)
- BSFC is a bowl: minimum at mid speed and part-to-full load; high at low speed (heat loss) and high speed (friction)

> 2024 Q5(a) 12 marks
>
> Define the following engine performance parameters: (i) Volumetric efficiency, (ii) Brake specific fuel consumption, (iii) Indicated mean effective pressure, (iv) Brake power.

> 2023 Q5(a) 14 marks
>
> What is the physical significance of volumetric efficiency? Draw and explain the following engine performance curves: (i) Brake torque, (ii) BSFC, (iii) volumetric efficiency.

2022 Q5(a) 12 marks

Discuss the following engine performance parameters with respect to engine speed using necessary diagrams: (i) Brake power, (ii) Brake torque, (iii) Brake specific fuel consumption.

2020 Q4(a) 9 marks

Discuss the following engine performance parameter with necessary diagrams: (i) Brake power (ii) Specific fuel consumption (iii) Volumetric efficiency.

Full answer: ESE 4105 Study Pack, topic B1.

B6 Valve timing diagram, valve overlap, why inlet valves are larger

Must study · Likely · 73% · Easy

Asked in 4 of 5 papers · 41 marks in total

STUDY THESE

- IVO 10-30° bTDC, IVC 30-60° aBDC, EVO 30-55° bBDC, EVC 10-20° aTDC (4-stroke petrol)
- Early/late opening because valves need time to open and gas has inertia: ram effect, blow-down, scavenging
- Overlap: both valves open near TDC; good at high speed, but fresh charge loss and backflow at low speed. Inlet valve larger because intake ΔP is small (~0.1 bar) vs exhaust pushed by cylinder pressure

2024 Q5(b) 10 marks

Briefly explain valve timing diagram and its functions of a typical four stroke engine with proper sketches.

2022 Q7(b) 14 marks

Why do engine inlet valves and exhaust valves open earlier and close later? Why are inlet valves made larger than exhaust valves?

2021 Q6(b) 12 marks

Discuss why actual valve timing differ from theoretical valve timing. Also, write the consequences of valve overlapping.

2020 Q1(c) 5 marks

Why are the intake valves larger than exhaust valves?

Full answer: ESE 4105 Study Pack, topic B6.

B10 Stages of combustion in SI engine

Must study · Likely · 73% · Easy

Asked in 3 of 5 papers · 32 marks in total

STUDY THESE

- I: ignition lag / flame development (A-B), ~10° CA, chemical, ~5-10 % burnt
- II: rapid flame propagation (B-C), 80-90 % burnt, steep pressure rise, peak ~10-15° aTDC
- III: after-burning (C-D) near walls during expansion; draw the p-θ firing vs motoring curve

> 2024 Q3(c) 10 marks
>
> How does combustion occur in SI engines? Illustrate and explain the different stages from ignition to complete burning.

> 2023 Q6(a) 10 marks
>
> Discuss the different stages of combustion in a typical SI engine with suitable diagrams.

> 2021 Q7(b) 12 marks
>
> Discuss the different stages of combustion in SI engine using appropriate diagrams.

Full answer: ESE 4105 Study Pack, topic B10.

B5 Stoichiometry, equivalence ratio, rich vs lean; why CI runs lean

Must study · Likely · 67% · Easy

Asked in 4 of 5 papers · 38 marks in total

STUDY THESE

- Stoichiometric = exactly enough O_2 for complete combustion to CO_2 and H_2O (A/F ≈ 14.7 petrol, 14.5 diesel)
- ϕ = (F/A)actual / (F/A)stoich = 1/λ; $\phi < 1$ lean, $\phi > 1$ rich
- CI is quality governed: load set by fuel quantity with full air, and needs excess air to find oxygen for the spray before soot forms → $\phi \approx 0.3$-0.8

> 2023 Q5(c) 9 marks
>
> What is meant by "Equivalence ratio"? "Generally a CI engine operates with lean mixture" – Explain.

> 2022 Q6(a) 11 marks
>
> What is meant by combustion stoichiometry? Distinguish fuel rich combustion from fuel lean combustion.

> 2021 Q5(a) 9 marks
>
> What is meant by "Equivalence ratio"? "Generally a CI engine operates with lean mixture" – Explain.

> 2020 Q4(b) 9 marks
>
> What is meant by fuel rich combustion and fuel lean combustion? (first part)

Full answer: ESE 4105 Study Pack, topic B5.

B12 Surface ignition, pre- vs post-ignition, end gas, flame front

Must study · Likely · 60% · Easy

Asked in 3 of 5 papers · 29 marks in total

STUDY THESE

- Surface ignition = mixture ignited by a hot surface (plug electrode, exhaust valve, deposit), not by the spark
- Pre-ignition (before spark) is worse: uncontrolled advance, compression work against burning gas, overheating, holes in piston; post-ignition (after spark) just speeds combustion
- End gas = unburnt mixture ahead of the flame front; flame front = thin reaction zone separating burnt and unburnt gas

> 2023 Q6(b) 7 marks
>
> What are meant by surface ignition? "Pre-ignition is more harmful than post-ignition" – Discuss.

> 2022 Q7(a) 11 marks
>
> What is meant by surface ignition? "Pre-ignition is more harmful than post-ignition" – Justify the statement.

> 2021 Q7(a) 11 marks
>
> What are meant by: (i) surface ignition, (ii) end gas, and (iii) flame front? "Pre-ignition is more harmful than post ignition" – Explain.

Full answer: ESE 4105 Study Pack, topic B12.

B13 Knock in SI vs CI; ignition delay SI vs CI; SI knock limited

Must study · Likely · 67% · Medium

Asked in 4 of 5 papers · 77 marks in total

STUDY THESE

- SI knock = auto-ignition of the end gas at the END of combustion; CI knock = too much fuel burning at once at the START (long delay)
- Factors that cure SI knock (low T, low p, long delay, low r) cause CI knock, and vice versa
- SI is knock-limited: knock caps compression ratio, boost and spark advance, hence efficiency; CI welcomes high T and r

> 2023 Q6(c) 12 marks
>
> How do ignition delay and knock in SI engine differ from those in CI engine? Explain with appropriate diagrams.

> 2022 Q7(c) 10 marks
>
> How knocking in SI engines differs from that in CI engines? Explain with appropriate diagrams.

> 2022 Q8(c) 13 marks
>
> What is meant by end gas auto-ignition? "SI engine performance is knock limited but CI engine not" – Justify the statement.

> 2021 Q8(a) 10 marks
>
> How knock in SI engine differs from knock in CI engine?

> 2021 Q7(c) 12 marks
>
> What is meant by "Knock" in relation to SI engine? "SI engine performance is knock limited but CI engine not" – justify the statement.

> 2020 Q5(b) 12 marks
>
> How ignition delay in SI engine differs from ignition delay in CI engine? "Factors tend to increase knock in SI engine reduce knock in CI engine" – Justify.

> 2020 Q5(c) 8 marks
>
> What is meant by abnormal combustion? Explain the origin of knock in SI engine with the help of auto-ignition theory.

Full answer: ESE 4105 Study Pack, topic B13.

B4 Indicator diagram: indicated and pumping work; pumping loss, WOT, turbocharged engines

Must study · Likely · 67% · Medium

Asked in 4 of 5 papers · 53 marks in total

STUDY THESE

- Gross indicated work = area of the compression-expansion loop; pumping work = area of the exhaust-intake loop (negative in NA engines)
- Net indicated work = gross − pumping; SI throttling makes intake pressure low → large pumping loss at part load; CI has no throttle
- At WOT throttle loss vanishes so pumping loss is minimum; turbocharged: intake pressure > exhaust pressure, so pumping work can be positive

> 2023 Q5(b) 6 marks
>
> With the help of indicator diagram, explain: (i) Net indicated work per cycle, (ii) Pumping work per cycle.

> 2023 Q5(d) 6 marks
>
> How does pumping work in a turbocharged engine differ from that in a naturally aspirated engine? Discuss.

> 2022 Q5(b) 13 marks
>
> What is meant by pumping loss? Why pumping loss is higher in conventional SI engine compared to CI engine? How to reduce it?

> 2021 Q5(c) 10 marks
>
> With the help of indicator diagram, explain: (i) Indicated work/cycle; (ii) Pumping work/cycle.

> 2021 Q5(d) 6 marks
>
> Why pumping loss is minimum for engine operating at WOT condition?

> 2020 Q4(c) 12 marks
>
> What is meant by "wide open throttle (WOT)" operation? Why pumping loss is higher in conventional SI engine compared to CI engine? How to reduce it?

Full answer: ESE 4105 Study Pack, topic B4.

CI knock factors and ignition delay; good SI fuel is bad

B14 CI fuel; poor fuel in large slow CI; smoke limited

Must study · Likely · 60% · Medium

Asked in 3 of 5 papers · 69 marks in total

STUDY THESE

- Delay ↓ (less knock) with higher cetane, compression ratio, intake T and p, load, speed (in ms), turbulence; injection timing near optimum
- Good SI fuel = high octane = high SIT = long delay → CI knock; good CI fuel = high cetane = low SIT → SI knock
- Large slow CI: long time per degree, so delay in degrees is short → can burn heavy low-cetane fuel. CI is smoke limited: more fuel at full load → soot before knock

> 2023 Q7(c) 13 marks
>
> Discuss the major factors that influence the tendency of knock in CI engine.

> 2023 Q7(b) 10 marks
>
> "A good SI engine fuel is a bad CI engine fuel" – Explain with appropriate diagrams.

> 2023 Q8(d) 5 marks
>
> Why poor ignition quality fuel can be used in low speed large sized CI engine? Explain.

> 2023 Q6(d) 6 marks
>
> "CI engine performance is smoke limited" – Justify the statement.

> 2022 Q6(b) 10 marks
>
> "CI engine performance is smoke limited" – Justify the statement.

> 2021 Q8(c) 10 marks
>
> Discuss the factors that affect ignition delay in CI engines.

> 2021 Q8(b) 10 marks
>
> "A good SI engine fuel is a bad CI engine fuel" – justify the statement.

> 2021 Q8(d) 5 marks
>
> Why low ignition quality fuel can be used in very large, slow-speed CI engine?

Full answer: ESE 4105 Study Pack, topic B14.

B9 SI mixture formation strategies; HCCI and DISI

Good to have · Likely · 60% · Medium

Asked in 2 of 5 papers · 21 marks in total

STUDY THESE

- Three strategies: carburetor, port/manifold fuel injection (single-point TBI, multipoint PFI), direct in-cylinder injection (GDI)
- HCCI: premixed lean charge auto-ignites by compression; low NOx and soot, high efficiency, but hard to control timing
- DISI (GDI): fuel sprayed into cylinder; stratified lean charge at part load, charge cooling allows higher r, but PM and injector deposits

> 2024 Q6(c) 9 marks
>
> Write short notes on: (i) Homogeneous charge compression ignition (HCCI), (ii) Direct injection spark ignition (DISI).

> 2023 Q7(a) 12 marks
>
> Discuss the three strategies of mixture formation process in SI engine with suitable diagrams.

Full answer: ESE 4105 Study Pack, topic B9.

B8 Scavenging; two-stroke vs four-stroke; why 2-stroke CI is common but 2-stroke SI is not

Must study · Possible · 40% · Easy

Asked in 3 of 5 papers · 51 marks in total

STUDY THESE

- Scavenging = pushing burnt gas out with fresh charge/air in a 2-stroke (cross, loop, uniflow)
- 2-stroke SI scavenges with fuel-air mixture → short-circuiting loses 20-30 % fuel and HC; 2-stroke CI scavenges with pure air → no fuel loss
- 4-stroke: better economy, less pollution, better lubrication; 2-stroke: more power per weight, simpler

> 2022 Q3(b) 10 marks
>
> What are the impacts of scavenging? Why scavenging requires more attention in case of 2-stroke engine than that for 4-stroke engine?

> 2022 Q8(b) 8 marks

> Why 2-stroke CI engines find wider use whereas 2-stroke SI engines are used only to a limited extent?

> 2021 Q6(a) 7 marks
> What is meant by scavenging? Why scavenging is more important for 2-stroke engine than for 4-stroke engine.

> 2021 Q6(d) 7 marks
> Why 2-stroke CI engine is more common than 2-stroke SI engine? (second part)

> 2020 Q4(b) 9 marks
> Why two stroke CI engines find wider use whereas two stroke SI engines are used only to a limited extent? (second part)

> 2020 Q2(b) 10 marks
> "Four stroke IC engine is economical and less pollutant than two stroke engine" – Justify the statement.

Full answer: ESE 4105 Study Pack, topic B8.

B11 Stages of combustion in CI engine (ROHR vs crank angle); flames; SIT

Must study · Possible · 53% · Medium

Asked in 3 of 5 papers · 45 marks in total

STUDY THESE

- Four stages: ignition delay (a-b), premixed / uncontrolled combustion (b-c, ROHR spike), mixing-controlled / diffusion combustion (c-d), late combustion (d-e)
- SIT = lowest temperature at which a fuel-air mixture ignites by itself without spark or flame
- Premixed flame: fuel and air mixed before, flame travels (SI, blue); diffusion flame: mixing at the flame, rate set by mixing (CI, yellow, soot)

> 2023 Q8(c) 11 marks
> Describe the different stages of combustion in a CI engine with the help of rate of heat release (ROHR) versus crank angle (θ) diagram.

> 2022 Q6(c) 14 marks
> Describe the different stages of combustion in CI engine with the help of rate of heat release (RDHR) verses crank angle (θ) diagram.

> 2022 Q5(c) 10 marks

What is the physical meaning of SIT? Explain the key characteristics of premixed flame and diffusion flame.

2020 Q5(a) 10 marks

How flames are classified? Describe the different stages of combustion in CI engine with the help of "rate of heat release-crank angle" diagram.

Full answer: ESE 4105 Study Pack, topic B11.

B3 Volumetric efficiency: significance, mean piston speed, valve timing and speed effects, quasi-static effect

Must study · Possible · 33% · Medium

Asked in 2 of 5 papers · 29 marks in total

STUDY THESE

- ηv measures breathing: power ∝ ηv, because fuel that can be burnt is limited by air inducted
- Mean piston speed 2LN is held at 8-15 m/s: above it inertia stress, friction, flow losses (choking at inlet valve) rise
- Late IVC uses ram effect at high speed but pushes charge back at low speed; quasi-static effects = fuel vapour partial pressure, p_e/p_i and r_c

2022 Q2(c) 10 marks

How quasi-static effect influences the volumetric efficiency of an internal combustion engine. (second part)

2021 Q5(b) 10 marks

What is the physical significance of volumetric efficiency? "Mean piston speed ranges within 8 to 15 m/s" – Discuss.

2021 Q6(c) 9 marks

How inlet valve opening and closing, and engine speed affect volumetric efficiency?

Full answer: ESE 4105 Study Pack, topic B3.

B2 Rope brake engine test numerical (BP, IP, ηm, ηbth, BSFC)

Skip if short · Possible · 33% · Medium

Asked in 1 of 5 papers · 20 marks in total

STUDY THESE

- BP = (W − S) π D N / 60
- p_mi = a s / l; IP = p_mi L A n / 60 with n = N/2 for four-stroke
- ηm = BP/IP; ηbth = BP × 3600 / (ṁ_f CV); BSFC = ṁ_f / BP

> 2024 Q6(b) 20 marks
>
> Single-cylinder four-stroke oil engine with rope brake: brake wheel diameter 600 mm, dead load 400 N, spring balance 50 N, speed 600 rpm, indicator diagram area 420 mm², length 60 mm, spring scale 1.1 bar/mm, bore 100 mm, stroke 150 mm, oil 0.815 kg/h, CV 44000 kJ/kg. Calculate BP, IP, mechanical efficiency, brake thermal efficiency and BSFC.

Full answer: ESE 4105 Study Pack, topic B2.

B7 Air-standard cycles: Otto efficiency derivation; Diesel cut-off ratio and γ

Good to have · Possible · 33% · Hard

Asked in 1 of 5 papers · 19 marks in total

STUDY THESE

- Otto: η = 1 − 1/r^(γ−1) from $T_2/T_1 = T_3/T_4 = r^{(\gamma-1)}$
- Diesel: η = 1 − [1/(γ r^(γ−1))] (ρ^γ − 1)/(ρ − 1); bigger cut-off ratio ρ → lower efficiency
- Higher γ → higher efficiency (air γ = 1.4; rich mixture / hot gas lower γ)

> 2024 Q5(c) 13 marks
>
> Sketch the P-V diagram of Otto cycle and derive mathematical expression for thermal efficiency of Otto cycle.

> 2024 Q6(a) 6 marks
>
> How does cut-off ratio affect diesel cycle efficiency? What role does the specific heat ratio (γ) play in cycle efficiency?

Full answer: ESE 4105 Study Pack, topic B7.

Abbreviations used in this pack

Every short form, written out. The answers also spell each one out the first time it appears.

Short form	Full meaning
A/F	air-fuel ratio
API	American Petroleum Institute (gravity scale)
BDC	bottom dead centre
BMEP	brake mean effective pressure
BP	brake power
BSFC	brake specific fuel consumption
CFR	Cooperative Fuel Research (standard test engine)
CI	compression ignition (diesel engine)
CN	cetane number
CNG	compressed natural gas
CO	carbon monoxide
CV	calorific value
DISI	direct injection spark ignition
DPF	diesel particulate filter
EGR	exhaust gas recirculation
EVC	exhaust valve closes
EVO	exhaust valve opens
F/A	fuel-air ratio
FAME	fatty acid methyl ester (biodiesel)
FBC	fluidised bed combustion
FC	fixed carbon
FP	friction power
GDI	gasoline direct injection
HC	hydrocarbons (unburnt fuel)
HCCI	homogeneous charge compression ignition
HCV	higher calorific value
IC	internal combustion

IMEP	indicated mean effective pressure
IP	indicated power
ISFC	indicated specific fuel consumption
IVC	inlet valve closes
IVO	inlet valve opens
LCV	lower calorific value
LPG	liquefied petroleum gas
MEP	mean effective pressure
NG	natural gas
NOx	nitrogen oxides ($NO + NO_2$)
ON	octane number
PCV	positive crankcase ventilation
PM	particulate matter
ROHR	rate of heat release
rpm	revolutions per minute
RVP	Reid vapour pressure
SAE	Society of Automotive Engineers (oil grade)
SCR	selective catalytic reduction
SI	spark ignition (petrol engine)
SIT	self-ignition temperature
SOF	soluble organic fraction
TDC	top dead centre
TWC	three-way catalytic converter
VM	volatile matter
WOT	wide open throttle

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